

Endeavor

OpenSCAP evidence to OSCAL

Start-to-finish milestone map

A review roadmap from v0.1 evidence conversion through a stable, publishable v1.0 release.

North star	Convert OpenSCAP OVAL and later XCCDF/ARF assessment evidence into valid, provenance-preserving OSCAL artifacts for assessment, authorization, continuous monitoring, and remediation workflows.
v1.0 promise	A stable CLI and mapping specification with schema-valid outputs, reproducible test corpus, evidence traceability, signed releases, and migration guidance.
Hard boundary	Endeavor does not reimplement OVAL/XCCDF evaluation, silently invent OSCAL control mappings, or execute remediation in its first major release.

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Roadmap at a glance

The sequence is intentionally gate-driven. Each phase yields an auditable artifact that reduces semantic, security, or release risk before the next layer expands scope.

Phase	Milestone	Primary outcome	Exit evidence
0	Foundation locked	v0.1 mapping and corpus contract	Reviewed contract and fixture manifest
1	Vertical slice	OVAL Results -> OSCAL Assessment Results CLI	Schema-valid golden outputs
2	Trustworthy beta	Hardened parser, deterministic evidence bundles	Security and regression gates
3	Usable alpha	Mappings, human review report, run diffs	User workflow validation
4	SCAP expansion	ARF and XCCDF result ingestion	Cross-format provenance tests
5	Governance workflow	Assessment Plans, POA&M; candidates, CI	End-to-end authorization demo
6	Release candidate	Compatibility matrix and operations readiness	Reproducible release rehearsal
7	v1.0	Stable public interface and support posture	Signed, documented GA release

Decision cadence

Review at every exit gate. Do not advance on feature completion alone: advance only when the stated evidence, provenance, safety, and usability signals are present.

Milestones 0 through 2 - establish semantic trust

0. Foundation locked

Purpose: Freeze the narrow v0.1 evidence-adapter contract before implementation.

Deliverables: Published mapping contract; namespace/property conventions; pinned OSCAL version; synthetic fixture manifest; license and source policy.

Exit gate: Maintainers agree that all six OVAL states are preserved and no control finding is invented without an explicit mapping.

Depends on: Existing discovery documents and v0.1 mapping contract.

1. v0.1 vertical slice

Purpose: Prove an end-to-end conversion for small, controlled OVAL Results inputs.

Deliverables: Language/runtime decision; `endeavor inspect` and `endeavor convert`; OVAL XML parser configured with external retrieval disabled; Assessment Results JSON emitter; SHA-256 evidence resources; deterministic IDs; five synthetic fixture pairs and golden outputs.

Exit gate: Every fixture converts noninteractively, validates against the pinned OSCAL schema, preserves source status and IDs, and produces no trusted output on malformed input.

Depends on: Phase 0 and schema-validated synthetic OVAL fixtures.

2. Trustworthy beta

Purpose: Make the vertical slice safe and repeatable enough for external technical evaluation.

Deliverables: Resource bounds for XML and archives; structured diagnostics and exit codes; redaction policy; unit, regression, fuzz, and determinism suites; SBOM and dependency policy; CI validation job.

Exit gate: Threat-focused tests cover entity expansion, unsafe paths/URIs, malformed XML, and sensitive log handling. A clean CI run validates both source and output artifacts.

Depends on: Phase 1 converter and corpus.

Phase 2 review question

Can a security engineer run the converter in CI without needing to trust opaque parsing, an undocumented status mapping, or uncontrolled evidence disclosure?

Milestones 3 through 5 - make the evidence operational

3. Usable alpha

Purpose: Add the workflows that make OSCAL output easier to inspect and compare than raw SCAP artifacts.

Deliverables: Versioned mapping-file format; unmapped-ID report; accessible semantic HTML report; `endeavor findings`; `endeavor diff`; CLI JSON output and stable stderr/exit-code contract.

Exit gate: A user can explain a changed result across two runs and see every mapped, unmapped, and source-evidence link without scraping HTML.

Depends on: Phase 2 quality gates and representative multi-run fixtures.

4. SCAP expansion

Purpose: Extend ingestion without weakening the evidence contract.

Deliverables: ARF/XCCDF result normalization; profile/tailoring preservation; target and evaluator metadata normalization; compatibility matrix for supported OpenSCAP and schema versions.

Exit gate: Cross-format tests demonstrate that source profile, tailoring, target, evaluator, and raw evidence remain traceable in output.

Depends on: Phase 3 mapping and report surface.

5. Governance workflow

Purpose: Connect assessment evidence to authorization and continuous-monitoring processes while retaining human approval boundaries.

Deliverables: Optional Assessment Plan generation; POA&M; candidate generation; signed evidence manifests; CI examples; reference integration with a system-security-plan or control catalog workflow.

Exit gate: A reproducible demo links a scan result to its plan, evidence resources, mappings, candidate remediation work, and a follow-up scan delta. No remediation is executed by Endeavor.

Depends on: Phase 4 cross-format ingestion and a versioned mapping specification.

Scope guardrail

POA&M; candidates are proposals. Endeavor must continue to distinguish generated recommendations, approved work, executed work, and verified remediation evidence.

Milestones 6 and 7 - release with durable trust

6. Release candidate

Purpose: Turn a capable tool into an operable open-source project.

Deliverables: CLI/API stability proposal; deprecation policy; installer/container strategy; release automation; signed artifacts and checksums; reproducible-build rehearsal; accessibility review; security disclosure process; contributor and maintainer documentation.

Exit gate: Independent clean-environment installation, corpus execution, schema validation, and artifact verification all pass. Release notes state supported versions and known limits.

Depends on: Phase 5 end-to-end workflow plus a completed compatibility matrix.

7. v1.0 general availability

Purpose: Publish a stable evidence-adaptor contract that users can adopt with confidence.

Deliverables: Versioned mapping specification; stable CLI/API; reproducible corpus; migration guides; supported-release policy; signed v1.0 tag, artifacts, SBOM, checksums, and accessible documentation.

Exit gate: Maintainers approve the release checklist and can reproduce the release. Public documentation clearly states the boundary, supported inputs, guarantees, non-goals, and disclosure path.

Depends on: Phase 6 release rehearsal and maintained issue triage process.

Cross-cutting tracks

Track	Run continuously from phase 1 onward	Proof at v1.0
Semantic fidelity	Versioned mappings, status-preservation tests, source artifact hashes, and unmapped reporting.	No undocumented lossy transformations in supported paths.
Security and privacy	Safe XML defaults, bounded resources, no network retrieval, redaction tests, dependency review.	Published security policy and release evidence.
Accessibility	Pipe-safe CLI, JSON output, semantic docs and reports, keyboard and screen-reader checks.	WCAG 2.2 AA release review for web surfaces.
Community	Issue templates, fixtures contribution rules, transparent support boundaries.	Contributor guide and sustainable triage policy.

Execution controls and immediate next work

The roadmap is complete when Endeavor can be installed cleanly, convert supported evidence into schema-valid OSCAL with a documented preservation contract, prove that behavior through a reproducible corpus, and publish signed, accessible, supportable artifacts.

Order	Next action	Concrete acceptance signal
1	Choose the implementation language and packaging strategy for the v0.1 CLI.	A minimal, reproducible developer setup and CI job can run locally and in GitHub Actions.
2	Author and schema-validate the synthetic OVAL Results/Definitions fixture pairs declared in the manifest.	Five credible, license-safe inputs exist with documented intended source statuses.
3	Implement the inspect/convert vertical slice and golden OSCAL validation.	Phase 1 exit gate passes automatically in CI.

Review inputs

This map is derived from Endeavor's product direction, OpenSCAP user experience inventory, OSCAL experience design, quality strategy, and v0.1 OVAL Results mapping contract in the repository. It is a milestone plan, not a calendar commitment; dates and staffing can be added after the v0.1 implementation stack is selected.